MLDL PRACTICAL 4

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AIM :
Implement K-Nearest Neighbors (KNN) and evaluate model performance

Dataset Source

- **Dataset Name:** ICC Cricket World Cup 2023
- **Dataset File:** `CWC2023.csv`
- **Source:** Experiment-provided dataset containing World Cup 2023 match statistics

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Dataset Description

The dataset contains match-level information collected during the ICC Cricket World Cup 2023.

- **Number of instances:** 48 matches
- **Number of features:** 44
- **Target Variable:** `Wining Team`
- **Problem Type:** Multiclass classification

Key Features

- Match details (date, city, stadium)
- Team details (Team 1, Team 2, Toss Winner)
- Match statistics (runs, wickets, overs, run rate)

Dataset Characteristics

- Mixed numerical and categorical attributes
- Small dataset size
- High dimensionality
- Requires encoding and normalization before applying KNN

Mathematical Foundation of KNN

K-Nearest Neighbors is a **non-parametric**, **instance-based**, and **lazy learning** algorithm.

Distance Metric (Euclidean Distance)

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Where:

- xxx is the test data point

- yyy is a training data point
- nnn is the number of features

Classification Rule

$$\hat{y} = \text{majority class among K nearest neighbors}$$

The class most frequently occurring among the **K nearest matches** determines the predicted winning team.

Methodology / Workflow

Steps Followed

1. Load the ICC Cricket World Cup 2023 dataset
2. Handle missing or inconsistent values
3. Encode categorical attributes such as team names and venues
4. Normalize numerical features to ensure equal distance contribution
5. Split the dataset into training and testing sets
6. Apply the KNN classifier
7. Predict match outcomes
8. Evaluate model performance

Workflow Representation

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Dataset → Preprocessing → Feature Scaling
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```
→ Train-Test Split → KNN Classification
→ Prediction → Performance Evaluation
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Performance Evaluation

Evaluation Metrics Used

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Performance Analysis

- KNN provides reasonable accuracy on the dataset
- Model performance is highly dependent on the choice of **K**
- Proper feature scaling significantly improves results
- Small dataset size can cause fluctuations in predictions

Hyperparameter Selection

Key Hyperparameter

- **K (Number of Neighbors)**

Impact of K

- Small K → Overfitting (high sensitivity to noise)
- Large K → Underfitting (loss of local patterns)
- Optimal K balances bias and variance

Limitations of KNN

- Computationally expensive during prediction
- Sensitive to irrelevant or noisy features
- Performance degrades with high dimensional data
- Not suitable for very large datasets
- Requires careful preprocessing and scaling

OUTPUT

	Match ID	Match Date	Match Time	City	Stadium	Team A	Team B	Toss Winner	Toss Decision	Score A
0	1	05-10-2023	2:00 PM	Ahmedabad	Narendra Modi Stadium	England	NewZealand	NewZealand	Field	282
1	2	06-10-2023	2:00 PM	Hyderabad	Eden Gardens	Pakistan	Netherlands	Netherlands	Field	286
2	3	07-10-2023	2:00 PM	Dharamshala	HPCA Stadium	Afghanistan	Bangladesh	Bangladesh	Field	156
3	4	07-10-2023	2:00 PM	Delhi	Arun Jaitley Stadium	South Africa	Sri Lanka	Sri Lanka	Field	428
4	5	08-10-2023	2:00 PM	Chennai	M. A. Chidambaram Stadium	Australia	India	Australia	Bat	199